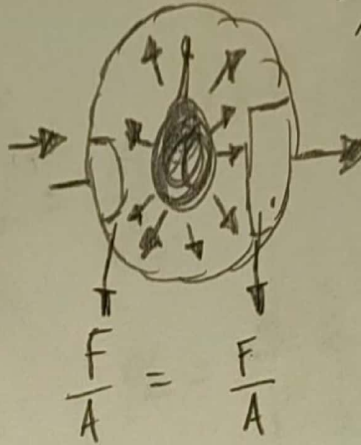
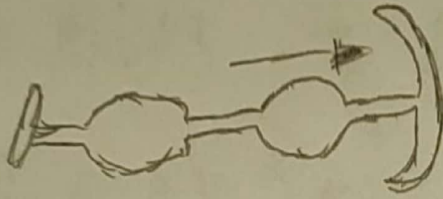


Segundo Examen Parcial

1)



$$\frac{F}{A}$$

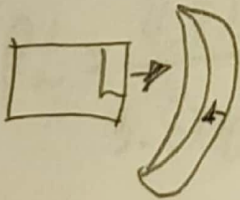
$$A_1 = 1.8 \text{ cm}^2$$

$$F = 44 \text{ N}$$

$$A_2 = 6.4 \text{ cm}^2$$

$$\frac{48}{0.018} = \frac{x}{0.004}$$

$$F_2 = \frac{44}{0.018} \cdot 0.064 = 156.44 \text{ N}$$



$$T = F \cdot d \cdot 0.34$$

$$F = M \cdot N$$

$$f_x = 0.5 \cdot 156.44$$

$$T = 0.5 \cdot 156.44 \cdot 0.34 = 26.59 \text{ N/m}$$

3 2)

$$mV = 1 \text{ Kg}$$

$$mI = 1.8 \text{ Kg}$$

$$Vb = 3.8 \times 10^{-3} \text{ m}^3$$

$$mE = 7.5 \text{ Kg}$$

$$mD = 3.5 \text{ Kg}$$

$$WLD = p. \text{ liquido desplazado}$$

$$MLD = m. \text{ liquido "}$$

$$VLD = Vol. \text{ " "}$$

$$WLD = MLD \cdot g \rightarrow \rho_L = \frac{ML}{VL}$$

$$WLD = \rho_L \cdot VLD \cdot g \quad \rho_L = \frac{MLD}{VLD}$$

Balanza E, muestra la masa total de mE del sistema =

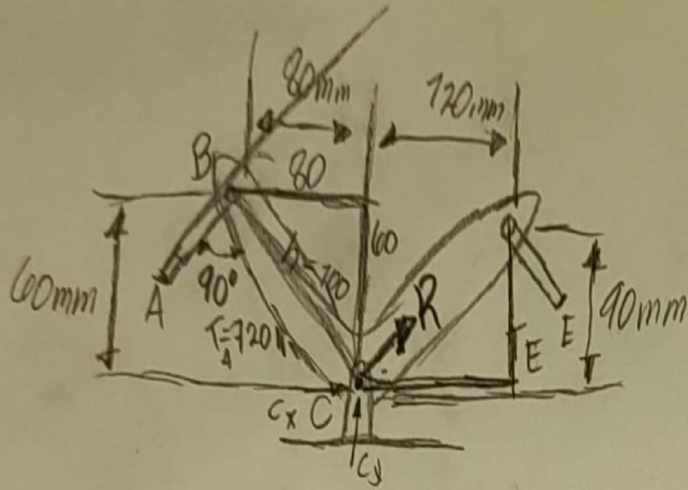
$$mE = mV + mI + MLD$$

$$MLD = mE - mV - mI$$

$$MLD = 7.5 \text{ Kg} - 1 \text{ Kg} - 1.8 \text{ Kg}$$

$$MLD = 4.7 \text{ Kg}$$

$$\rho_L = \frac{4.7 \text{ Kg}}{3.8 \times 10^{-3} \text{ m}^3} = 1236.842105 \text{ Kg/m}^3$$



$\Sigma M_C = 0$
 $720 \times 0.6 = 720 \times 0.6$
 $T = 600\text{ N}$

$$T_x = \sin 36.87 (720\text{ N}) = 432\text{ N}$$

$$T_y = \cos 36.87 (720\text{ N}) = 576\text{ N}$$

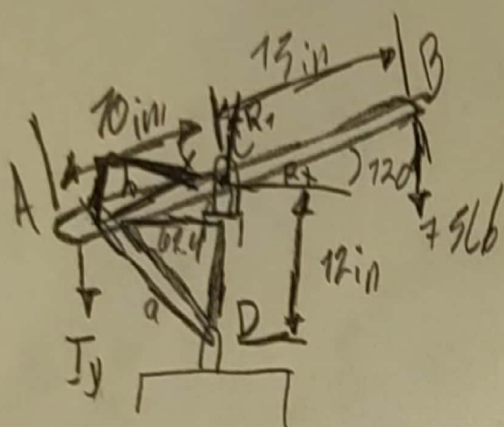
$$\Sigma F_y = R_y - 570\text{ N} - F_E = 0$$

$$R_y = 1176\text{ N}$$

$$R = \sqrt{(432)^2 + (1176)^2}$$

$$R = 1252.84\text{ N}$$

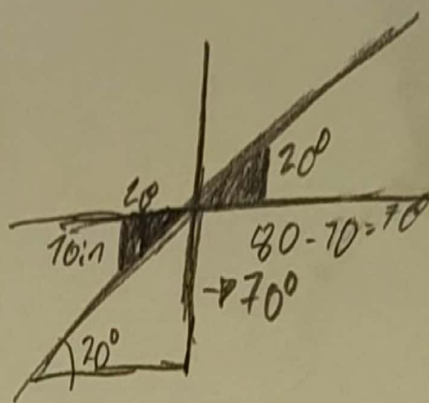
4)



$$b = 10 \text{ in}$$

$$c = 12 \text{ in}$$

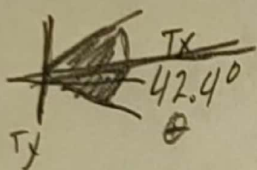
$$W =$$



$$a^2 = b^2 + c^2 - 2bc \cos \theta$$

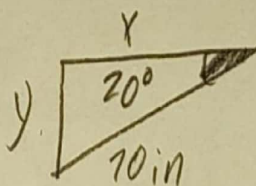
$$a^2 = 10^2 + 12^2 - 2(10)(12) \cos 70^\circ$$

$$a = 12.72 \text{ in}$$



$$T_x = \cos 42.4^\circ (T) = 88.08$$

$$T_y = \sin 42.4^\circ (T) = 80.9266$$



$$x = \cos 20^\circ = 9.4 \text{ in}$$

$$y = \sin 20^\circ = 3.42 \text{ in}$$

$$\sum F_x = R_x + T_x = 0$$

$$R_x = 88.08$$

$$\sum F_y = R_y - 75 \text{ lb} - T_y = 0$$

$$R_y = 115.42$$

$$\sum \tau = T_y(x) + T_x(y) - 75(b \cos \frac{20^\circ}{13 \text{ in}}) = 0$$

$$\sin 42.4^\circ (9.4) + \cos 42.4^\circ (T)(3.42) = 1057.75$$

$$8.86T = 1057.75$$

$$T = 119.27 \text{ lb}$$

$$5) \quad D = 12 \text{ m} \quad P_{He} = \frac{1}{7} \text{ pie} \quad m = 170 \text{ Kg}$$

$$r = 6 \text{ m}$$

$$V_{\text{globo}} = \frac{4\pi r^3}{3} = \frac{4\pi (6)^3}{3} = 904.8 \text{ m}^3$$

$$F_B = \rho_{\text{aire}} \cdot g \cdot V_{\text{globo}} = (1.16)(9.8)(904.8) = 10285.77$$

Total masa

$$M_{He} = P_{He} \cdot V = \left(\frac{1.16}{7}\right)(904.8) = 149.9 \text{ Kg}$$

$$M_{\text{total}} = M_{He} + M_p = 149.9 + 2185 = 319.9 \text{ Kg}$$

$$W = M_{\text{total}} \cdot g = 319.9 \cdot 9.81 = 3138. \text{ N} \quad 2 \text{ Kg}$$

$$F_{\text{net}} = F_B - W = 10285.77 - 3138 = 7147.77$$

$$F = 1779 \quad a = \frac{F}{m} = \frac{7147.77}{319.9} = \underline{22.35 \text{ m/s}^2}$$